

Exposé VIII

GROUPS OF CLASSES OF ABELIAN AND TRIANGULATED CATEGORIES. PERFECT COMPLEXES.

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In the present exposé, we quickly outline the notions and results which will be indispensable to us in the following exposés, concerning in particular pseudo-coherent and perfect complexes. A more complete exposé on the latter, with more detailed proofs, will appear in the first exposés of the seminar on the Riemann-Roch theorem for schemes, which will follow the present seminar.¹

1. The case of abelian categories

Let $\mathcal{C}$ be an abelian category and let $\mathcal{D}$ be a full subcategory of $\mathcal{C}$. A function f from $\mathcal{D}$ to an abelian group G is said to be additive if, for every exact sequence in $\mathcal{C}$

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0,$$

such that E' , E'' , and E are objects of $\mathcal{D}$, one has

$$f(E) = f(E') + f(E'').$$

Let f be an additive function from $\mathcal{D}$ to an abelian group G . The following properties are easy consequences of the definition:

$$\alpha) \quad f(0) = 0, \quad \beta) \quad f(X) = f(Y) \quad \text{if } X \text{ and } Y \text{ are isomorphic,}$$

$$\gamma) \quad f(X \oplus Y) = f(X) + f(Y).$$

$$\delta) \quad \text{Let } E = (E^i, d^i) \text{ be a finite complex in } \mathcal{D}$$

such that the objects $B^{i+1} = \text{Im}(d^i)$, $Z^i = \text{Ker}(d^i)$ and $H^i(E) = Z^i/B^i$ are in $\mathcal{D}$. Then

$$\sum_i (-1)^i f(E^i) = \sum_i (-1)^i f(H^i(E)).$$

$$\varepsilon) \quad \text{If } F \in \text{ob } \mathcal{D} \text{ has a finite filtration}$$

$$0 = F_0 \subset F_1 \subset \cdots \subset F_n = F,$$

and if one assumes that, for $0 < i \leq n$, F_i and F_i/F_{i-1} are objects of $\mathcal{D}$, then

$$f(F) = \sum_{i=1}^n f(F_i/F_{i-1}).$$

$$\psi) \quad \text{If } \mathcal{D} \text{ is stable under infinite direct sums, then } f = 0.$$

¹SGA 6 I

To prove δ), one uses the exact sequences

$$0 \longrightarrow Z^i \longrightarrow E^i \longrightarrow B^{i+1} \longrightarrow 0,$$

$$0 \longrightarrow B^i \longrightarrow Z^i \longrightarrow H^i(E) \longrightarrow 0.$$

The proof of ε) is by induction on n , using the exact sequence

$$0 \longrightarrow F_{n-1} \longrightarrow F \longrightarrow F/F_{n-1} \longrightarrow 0.$$

To prove ψ), let $E \in \text{ob } \mathcal{D}$ and

$$F = \bigoplus_{n \in \mathbb{Z}^+} F_n, \quad F_n = E, \quad n \in \mathbb{Z}^+.$$

Then $E \oplus F \simeq F$, and applying γ) gives

$$f(F) = f(F) + f(E),$$

hence $f(E) = 0$.

If, for every abelian group G , one considers the abelian group $A_{\mathcal{D}}(G)$ of additive functions from $\mathcal{D}$ to G , one obtains in an evident way a covariant functor

$$A_{\mathcal{D}} : \text{Ab} \longrightarrow \text{Ab}.$$

The functor $A_{\mathcal{D}}$ is representable. A representing pair $(K_{\mathcal{D}}(\mathcal{C}), \varphi)$ may be obtained as follows. Let L be the free abelian group generated by the objects of $\mathcal{D}$, and let R be the subgroup of L generated by the elements of the form $E - E' - E''$ for each exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

of $\mathcal{C}$. Put

$$K_{\mathcal{D}}(\mathcal{C}) = L/R.$$

The map φ is obtained by composing the two canonical maps

$$\text{ob } \mathcal{D} \longrightarrow L \longrightarrow L/R.$$

When there is no danger of confusion, one may abbreviate the notation:

$$K_{\mathcal{D}}(\mathcal{C}) = K(\mathcal{D}).$$

2. The case of triangulated categories

Let $\mathcal{C}$ be a triangulated category and let G be an abelian group. A function f from $\mathcal{C}$ to G is said to be additive if, for every distinguished triangle

$$\begin{array}{ccc} & Z & \\ w \swarrow & & \nwarrow v \\ X & \xrightarrow{u} & Y \end{array}$$

one has

$$f(Y) = f(X) + f(Z).$$

Given an additive function f , the following relations are immediate consequences of the definition:

- a) $f(0) = 0$, b) $f(X[n]) = (-1)^n f(X)$,
c) $f(X) = f(Y)$ if X and Y are isomorphic, d) $f(X \oplus Y) = f(X) + f(Y)$.

Indeed the following triangles are distinguished:

$$\begin{array}{ccc} & 0 & \\ \swarrow & & \searrow \\ X & \xrightarrow{1_X} & X \end{array} \quad \begin{array}{ccc} & X[1] & \\ \swarrow & & \searrow \\ X & \xrightarrow{\quad} & 0 \end{array}$$

$$\begin{array}{ccc} & 0 & \\ \swarrow & & \searrow \\ X & \xrightarrow{u} & Y \end{array} \quad \begin{array}{ccc} & Y[1] & \\ \swarrow & & \searrow \\ X \oplus Y & \xrightarrow{\quad} & X \end{array}$$

(u an isomorphism). To form the last triangle, we used the fact that the sum of two distinguished triangles is distinguished.

As in the case of abelian categories, one obtains in an evident way a functor

$$A : \mathbf{Ab} \longrightarrow \mathbf{Ab}$$

which associates to each abelian group G the abelian group $A(G)$ of additive functions on $\mathcal{C}$ with values in G . This functor is representable, and it is easy to construct a representing pair $(K(\mathcal{C}), \text{cl}_{\mathcal{C}})$. Namely, one first considers the free abelian group $\mathbb{Z}^{(\text{ob } \mathcal{C})}$ generated by the objects of $\mathcal{C}$, and then the subgroup R generated by the elements of the form $X - Y - Z$, where

$$\begin{array}{ccc} & Z & \\ \swarrow & & \searrow \\ Y & \xrightarrow{\quad} & X \end{array}$$

is a distinguished triangle. The quotient group $\mathbb{Z}^{(\text{ob } \mathcal{C})}/R$ is $K(\mathcal{C})$, and $\text{cl}_{\mathcal{C}}$ is obtained by composing the two canonical maps

$$\text{ob } \mathcal{C} \longrightarrow \mathbb{Z}^{(\text{ob } \mathcal{C})} \longrightarrow K(\mathcal{C}).$$

3. Functorial character

Let $\mathcal{C}$ and $\mathcal{C}'$ be two triangulated categories, and let $T : \mathcal{C} \rightarrow \mathcal{C}'$ be an exact functor from $\mathcal{C}$ to $\mathcal{C}'$, that is, an additive graded functor carrying distinguished triangles to distinguished triangles. Denote, as above, by

$$A, A' : \mathbf{Ab} \longrightarrow \mathbf{Ab}$$

the functors associated with the categories $\mathcal{C}$ and $\mathcal{C}'$. There results a functorial morphism $A' \rightarrow A$ given by composition with T , hence a homomorphism

$$K(T) : K(\mathcal{C}) \longrightarrow K(\mathcal{C}')$$

uniquely determined by the commutativity of the diagram

$$\begin{array}{ccc} \text{ob } \mathcal{C} & \xrightarrow{T} & \text{ob } \mathcal{C}' \\ \text{cl}_{\mathcal{C}} \downarrow & & \downarrow \text{cl}_{\mathcal{C}'} \\ K(\mathcal{C}) & \xrightarrow{K(T)} & K(\mathcal{C}'). \end{array}$$

It follows that, if the functors $T, T_1 : \mathcal{C} \rightarrow \mathcal{C}'$ are isomorphic, then $K(T) = K(T_1)$. In particular, if T is an equivalence of categories, then $K(T)$ is an isomorphism.

If $\mathcal{C}''$ is a third triangulated category and

$$F : \mathcal{C} \times \mathcal{C}' \longrightarrow \mathcal{C}''$$

is a bifunctor exact in each variable, then it is clear that there results a bilinear map

$$K(F) : K(\mathcal{C}) \times K(\mathcal{C}') \longrightarrow K(\mathcal{C}'')$$

such that the following relation holds:

$$K(F)(\text{cl}_{\mathcal{C}}(X), \text{cl}_{\mathcal{C}'}(X')) = \text{cl}_{\mathcal{C}''}(F(X, X')).$$

Proposition 3.1. Let $\mathcal{A}$ be a triangulated category, $\mathcal{B}$ a thick triangulated subcategory of $\mathcal{A}$, $\mathcal{A}/\mathcal{B}$ the quotient category of $\mathcal{A}$ by $\mathcal{B}$, $Q : \mathcal{A} \rightarrow \mathcal{A}/\mathcal{B}$ the canonical quotient functor, and $i : \mathcal{B} \rightarrow \mathcal{A}$ the inclusion functor. Then the following sequence is exact:

$$K(\mathcal{B}) \xrightarrow{K(i)} K(\mathcal{A}) \xrightarrow{K(Q)} K(\mathcal{A}/\mathcal{B}) \longrightarrow 0.$$

Proof. It is clear that there is a natural homomorphism

$$\text{Coker}(K(i)) \xrightarrow{\alpha} K(\mathcal{A}/\mathcal{B}).$$

On the other hand, there is an evident map

$$\text{ob}(\mathcal{A}/\mathcal{B}) \xrightarrow{\gamma} \text{Coker}(K(i)),$$

since $\text{ob}(\mathcal{A}/\mathcal{B}) = \text{ob}(\mathcal{A})$. Let $X, Y \in \text{ob}(\mathcal{A}/\mathcal{B})$ and suppose that they are isomorphic. We want to show that $\gamma(X) = \gamma(Y)$. The category $\mathcal{A}/\mathcal{B}$ is obtained by a calculus of fractions on the right or on the left, the multiplicative system S being formed by the morphisms f which are contained in a distinguished triangle (X', Y', Z', f, g, h) in which Z' is an object of $\mathcal{B}$. (S is a saturated multiplicative system.) It follows that there exist X_1 and morphisms

$$X_1 \xrightarrow{s} X, \quad X_1 \xrightarrow{t} Y$$

such that $s, t \in S$. It is therefore enough to prove that $\gamma(X) = \gamma(X_1)$, and this follows from the fact that there is a distinguished triangle of the form

$$(X_1, X, Z, s, \dots),$$

with $Z \in \text{ob } \mathcal{B}$.

The map γ is an additive function. Indeed, if (X, Y, Z, f, g, h) is a distinguished triangle of $\mathcal{A}/\mathcal{B}$, one must prove that

$$\gamma(X) + \gamma(Z) = \gamma(Y).$$

But the relation now follows because the triangle (X, Y, Z, f, g, h) is isomorphic to a distinguished triangle of $\mathcal{A}$. Consequently one obtains

$$\beta : K(\mathcal{A}/\mathcal{B}) \longrightarrow \text{Coker}(K(i)),$$

and it is immediate to check that β is the inverse of α .

4. Comparison with the case of abelian categories

Let $\mathcal{C}$ be an abelian category and $D(\mathcal{C})$ the derived category of $\mathcal{C}$. As usual, denote by $D^b(\mathcal{C})$ the full subcategory of $D(\mathcal{C})$ generated by the bounded complexes. By associating with every object X of $\mathcal{C}$ the complex whose components are all zero except the component in dimension zero, which is equal to X , one obtains a homomorphism

$$K(\mathcal{C}) \xrightarrow{\mu} K(D^b(\mathcal{C})).$$

This homomorphism is in fact an isomorphism. Indeed, we may use Proposition 3.1 to construct an inverse to μ , taking into account that

$$D^b(\mathcal{C}) \simeq K^{b,b}(\mathcal{C})/K^{b,\phi}(\mathcal{C}),$$

where $K^{b,\phi}(\mathcal{C})$ is the thick subcategory of $K^{b,b}(\mathcal{C})$ consisting of the acyclic complexes. Let $X^\bullet = (X_i) \in \text{ob}(K^{b,b}(\mathcal{C}))$. If one sets

$$f(X^\bullet) = \sum_i (-1)^i \text{cl}_{\mathcal{C}}(X_i),$$

one obtains an additive function from $K^{b,b}(\mathcal{C})$ with values in $K(\mathcal{C})$. Indeed, one first observes that, if $X^\bullet$ and $Y^\bullet$ are homotopic, then $f(X^\bullet) = f(Y^\bullet)$, since $f(X^\bullet) = f(H(X^\bullet))$. It follows that there is a homomorphism

$$K(D^b(\mathcal{C})) \xrightarrow{\nu} K(\mathcal{C}),$$

and it is immediate to check that it is the inverse of μ .

Since the invariant $K(\mathcal{C})$ is often of no interest when $\mathcal{C}$ is too large, it is useful to relativize it with respect to a subcategory $\mathcal{C}_o$ of $\mathcal{C}$. Thus let $\mathcal{C}_o$ be a full subcategory of $\mathcal{C}$. We may define, on the one hand, the group

$$K_o(\mathcal{C}) = K_{\mathcal{C}_o}(\mathcal{C}).$$

Suppose, on the other hand, that the following condition is satisfied:

$$(A) \quad \mathcal{C}_o \text{ is stable under finite direct sums.}$$

It is then easy to define the triangulated category $D_o^b(\mathcal{C})$ by using complexes in $\mathcal{C}$ with bounded cohomology and with components in the subcategory $\mathcal{C}_o$, and passing to the quotient by quasi-isomorphisms of such complexes. As above, there results a homomorphism

$$(*) \quad K_o(\mathcal{C}) \longrightarrow K(D_o^b(\mathcal{C})).$$

Suppose that the subcategory $\mathcal{C}_o$ also satisfies the following condition (B):

$$(B) \quad \text{If } X, Y \in \text{ob } \mathcal{C}_o \text{ and } X \xrightarrow{u} Y \text{ is an epimorphism in } \mathcal{C},$$

then $\text{Ker } u \in \text{ob } \mathcal{C}_o$. Reasoning as above, one can show that the homomorphism $(*)$ is an isomorphism.

The canonical inclusion functor

$$D_o^b(\mathcal{C}) \longrightarrow D^b(\mathcal{C})$$

is not fully faithful in general. However:

Proposition 4.1. Suppose that the subcategory $\mathcal{C}_o$ satisfies conditions (A), (B), and the following condition (C):

$$(C) \quad \text{Every diagram} \quad \begin{array}{ccc} A & \xrightarrow{u} & B \\ & & \uparrow v \\ & & L \end{array}$$

where u is an epimorphism and $L \in \text{ob } \mathcal{C}_o$, can be completed to a commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{u} & B \\ \uparrow & & \uparrow v \\ L_1 & \xrightarrow{w} & L \end{array}$$

such that $L_1 \in \text{ob } \mathcal{C}_o$ and w is an epimorphism. Under these conditions, the canonical functors

$$D_o^b(\mathcal{C}) \longrightarrow D^b(\mathcal{C}), \quad D_o^-(\mathcal{C}) \longrightarrow D^-(\mathcal{C})$$

are fully faithful. Moreover, the essential image of $D_o^b(\mathcal{C})$ (resp. $D_o^-(\mathcal{C})$) is stable under direct factors.

The proof of this proposition essentially uses a criterion of Verdier ([2], 4–2 Theorem) and will be given in the first exposé of the seminar on Riemann-Roch (SGA 6 I 2.7).

Corollary. Suppose, in addition, that the subcategory $\mathcal{C}_o$ is stable under Ker and Coker, hence is an abelian category, and denote by $D^b(\mathcal{C})_o$ the full triangulated subcategory of $D^b(\mathcal{C})$ generated by the complexes of $D^b(\mathcal{C})$ whose cohomology objects are isomorphic to objects of $\mathcal{C}_o$. Then the inclusion functor

$$D^b(\mathcal{C}_o) \longrightarrow D^b(\mathcal{C})_o$$

is an equivalence between the two categories, and consequently one obtains the sequence of isomorphisms

$$K(\mathcal{C}_o) \simeq K_o(\mathcal{C}) \simeq K(D^b(\mathcal{C}_o)) \simeq K(D^b(\mathcal{C})_o). \quad (*)$$

Observation. Suppose that the subcategory $\mathcal{C}_o$ satisfies conditions (A) and (B), and that every object of $\mathcal{C}$ admits a finite resolution on the left by elements of $\mathcal{C}_o$. In this case, $\mathcal{C}_o$ also satisfies condition (C), and the canonical functor

$$D_o^b(\mathcal{C}) \longrightarrow D^b(\mathcal{C})$$

is in fact an equivalence of categories. Consequently, one has the isomorphism

$$K_{\mathcal{C}_o}(\mathcal{C}) \simeq K(\mathcal{C}).$$

5. Pseudo-coherent complexes

Let $\mathcal{C}$ be an abelian category and $\mathcal{C}_o$ a full subcategory of $\mathcal{C}$ satisfying conditions (A), (B), (C). We shall denote by $D_o^-(\mathcal{C})_{\text{coh}}$ the essential image of $D_o^-(\mathcal{C})$ under the canonical functor

$$D_o^-(\mathcal{C}) \longrightarrow D(\mathcal{C}).$$

The objects of $D_o^-(\mathcal{C})_{\text{coh}}$ will be called $\mathcal{C}_o$ -pseudo-coherent complexes. Thus a complex of $\mathcal{C}$ is $\mathcal{C}_o$ -pseudo-coherent if it is isomorphic in $D(\mathcal{C})$ to a bounded-above complex whose components are objects of $\mathcal{C}_o$.

The category $D_o^-(\mathcal{C})_{\text{coh}}$ is a triangulated subcategory of $D(\mathcal{C})$, in the sense that every distinguished triangle of $D(\mathcal{C})$ with two vertices in the subcategory $D_o^-(\mathcal{C})_{\text{coh}}$ is isomorphic to a triangle whose three objects are in the subcategory $D_o^-(\mathcal{C})_{\text{coh}}$.

First, it is clear that the subcategory $D_o^-(\mathcal{C})_{\text{coh}}$ is stable under the translation automorphism $T : D(\mathcal{C}) \rightarrow D(\mathcal{C})$. Therefore, to prove that $D_o^-(\mathcal{C})_{\text{coh}}$ is a triangulated subcategory of $D(\mathcal{C})$, it is enough to check that if $U^\bullet, V^\bullet$ are objects of $D_o^-(\mathcal{C})_{\text{coh}}$ and if the triangle

$$\begin{array}{ccc} & W^\bullet & \\ w \swarrow & & \nwarrow v \\ U^\bullet & \xrightarrow{u} & V^\bullet \end{array}$$

is distinguished for the triangulated category structure of $D(\mathcal{C})$, then $W^\bullet$ is also an object of $D_o^-(\mathcal{C})_{\text{coh}}$. By hypothesis, there exist complexes $X^\bullet, Y^\bullet$, isomorphic in $D(\mathcal{C})$ to $U^\bullet$ and $V^\bullet$ respectively, whose components are objects of $\mathcal{C}_o$. We obtain a commutative square in $D(\mathcal{C})$:

$$\begin{array}{ccc} U^\bullet & \xrightarrow{u} & V^\bullet \\ \sim \downarrow & & \downarrow \sim \\ X^\bullet & \xrightarrow{f} & Y^\bullet \end{array}$$

But, after replacing $X^\bullet$ by another isomorphic complex, we may suppose, using the fact that the inclusion functor

$$D_o^-(\mathcal{C}) \longrightarrow D^-(\mathcal{C})$$

is fully faithful (Prop. 4.1), that f is an actual morphism of complexes. Let $Z^\bullet$ be the mapping cylinder of f , and let $g : Y^\bullet \rightarrow Z^\bullet$ be the canonical morphism. By the construction itself, the components of $Z^\bullet$ are objects of $\mathcal{C}_o$ (condition (A)). On the other hand,

$$\begin{array}{ccc} & Z^\bullet & \\ h \swarrow & & \nwarrow g \\ X^\bullet & \xrightarrow{f} & Y^\bullet \end{array}$$

being a distinguished triangle in $D(\mathcal{C})$, it follows that $Z^\bullet$ is isomorphic to $W^\bullet$.

The category $D_o^-(\mathcal{C})_{\text{coh}}$ is therefore endowed with a structure of triangulated category: a triangle in $D_o^-(\mathcal{C})_{\text{coh}}$ is distinguished if and only if it is distinguished in the triangulated category $D(\mathcal{C})$.

The intersection of $D_o^-(\mathcal{C})_{\text{coh}}$ with $D^b(\mathcal{C})_{\text{coh}}$ will be denoted by $D_o^b(\mathcal{C})_{\text{coh}}$.

An object M of $\mathcal{C}$ will be called pseudo-coherent if it is so as a complex reduced to degree zero; one proves that it is necessary and sufficient for this that it admit a resolution on the left whose components are objects of $\mathcal{C}_o$.

Proposition 5.1. A complex X is $\mathcal{C}_o$ -pseudo-coherent if and only if it is isomorphic to a complex which is bounded on the right and whose components are $\mathcal{C}_o$ -pseudo-coherent.

The proof of this proposition will appear in the first exposé of the seminar on Riemann-Roch (SGA 6 I 2.7).

6. Perfect complexes

An object of the category $D(\mathcal{C})$ will be called a $\mathcal{C}_o$ -perfect complex if it is isomorphic to an object of $D_o^b(\mathcal{C})$, that is, to a bounded complex all of whose components are in $\mathcal{C}_o$. We shall denote by $D_o(\mathcal{C})_{\text{parf}}$ the full subcategory of $D(\mathcal{C})$ generated by the perfect complexes.

Thus the category $D_o(\mathcal{C})_{\text{parf}}$ is nothing other than the essential image of $D_o^b(\mathcal{C})$ under the inclusion functor

$$D_o^b(\mathcal{C}) \longrightarrow D(\mathcal{C}).$$

It is easy to see that $D_o(\mathcal{C})_{\text{parf}}$ is a triangulated subcategory of $D(\mathcal{C})$, and hence is endowed with a structure of triangulated category.

7. An important special case

Let A be a ring and let $\mathcal{C}$ be the category of A -modules, on the left for example. Denote by $\mathcal{C}_o$ the full subcategory of $\mathcal{C}$ generated by the projective A -modules of finite type. The subcategory $\mathcal{C}_o$ satisfies conditions (A), (B), (C). In this case we use the following notation:

$$D^-(A)_{\text{coh}} = D_o^-(\mathcal{C})_{\text{coh}}, \quad D^b(A)_{\text{coh}} = D_o^b(\mathcal{C})_{\text{coh}},$$

$$D(A)_{\text{parf}} = D_o(\mathcal{C})_{\text{parf}}.$$

The complexes of $D(\mathcal{C})$ which are $\mathcal{C}_o$ -pseudo-coherent (resp. $\mathcal{C}_o$ -perfect) will simply be called pseudo-coherent (resp. perfect) complexes.

It will be useful to consider the following groups:

$$K_{\bullet}(A) = K(D^b(A)_{\text{coh}}),$$

$$K'(A) = K(D(A)_{\text{parf}}).$$

As for the group $K'(A)$, we have, by Proposition 4.1 and its corollary,

$$K'(A) = K(D(A)_{\text{parf}}) \simeq K(D_o^b(\mathcal{C})) \simeq K_{\mathcal{C}_o}(\mathcal{C}),$$

which shows that it is also the classical group of classes of projective modules of finite type over A .

In the special case where A is noetherian, the group $K_{\bullet}(A)$ also admits a classical description. For this, let $\mathcal{C}_1$ be the abelian category of modules of finite type. Then

$$K_{\bullet}(A) \simeq K(\mathcal{C}_1).$$

Indeed, using Proposition 4.1, one has

$$K(\mathcal{C}_1) \simeq K(D^b(\mathcal{C}_1)) \simeq K(D_{\mathcal{C}_1}^b(\mathcal{C})),$$

because $D^b(\mathcal{C}_1) \simeq D_{\mathcal{C}_1}^b(\mathcal{C})$ immediately.

In fact, if A is noetherian and $L^\bullet \in D^-(A)$, then it follows from Proposition 5.1 that $L^\bullet$ is pseudo-coherent if and only if the $H^i(L^\bullet)$ are of finite type.

For an arbitrary ring A , one has a homomorphism

$$K'(A) \xrightarrow{\gamma_A} K_\bullet(A),$$

which comes from the inclusion functor

$$(*) \quad D(A)_{\text{parf}} \longrightarrow D^b(A)_{\text{coh}}.$$

In general, the homomorphism γ_A is neither injective nor surjective. In the special case of a regular ring A , every A -module M of finite type has finite projective dimension and it follows, thanks to 5.1, that the inclusion functor $(*)$ is in fact an equivalence between the two categories. In that case, γ_A is therefore an isomorphism.

8. Tor of complexes

Let A be a not necessarily commutative ring, and let ${}_A\mathcal{C}$ (resp. $\mathcal{C}_A$) be the abelian category of left (resp. right) A -modules. By $K^-({}_A\mathcal{C})$ we shall denote the triangulated category of complexes of ${}_A\mathcal{C}$ bounded above, and by ${}_A\mathcal{D}$ the full triangulated subcategory of $K^-({}_A\mathcal{C})$ consisting of complexes whose components are flat left A -modules.

Let $X^\bullet$ be an object of $K^-({}_A\mathcal{C})$, and consider the functor

$$\Phi_{X^\bullet} = X^\bullet \otimes_A \cdot : K^-({}_A\mathcal{C}) \longrightarrow K^-(\text{Ab}).$$

Under these conditions, the following propositions hold.

1) Every acyclic object of ${}_A\mathcal{D}$ is transformed by $\Phi_{X^\bullet}$ into an acyclic object of $K^-(\text{Ab})$. Indeed, if $N^\bullet$ is an object of ${}_A\mathcal{D}$ such that $Z^i(N^\bullet)$, $B^i(N^\bullet)$, and $H^i(N^\bullet)$ are flat, one has the classical isomorphism

$$\bigoplus_{p+q=n} H_p(X^\bullet) \otimes H_q(N^\bullet) \longrightarrow H_n(X^\bullet \otimes_A N^\bullet).$$

2) Every object $Y^\bullet$ of $K^-({}_A\mathcal{C})$ is the target of a quasi-isomorphism whose source is an object of ${}_A\mathcal{D}$. It is enough for this to take a Cartan-Eilenberg resolution. We can therefore apply Theorem 2 of §2, no. 2 of [2] to conclude that $\Phi_{X^\bullet}$ admits a left derived functor:

$$\mathbf{L}\Phi_{X^\bullet} : D^-({}_A\mathcal{C}) \longrightarrow D^-(\text{Ab}).$$

The object $\mathbf{L}\Phi_{X^\bullet}(Y^\bullet)$ may be obtained, up to isomorphism, as follows. Let $P^\bullet$ be an object of ${}_A\mathcal{D}$ and let $P^\bullet \rightarrow Y^\bullet$ be a quasi-isomorphism. Then $\mathbf{L}\Phi_{X^\bullet}(Y^\bullet)$ is isomorphic to

$$Q(X^\bullet \otimes_A P^\bullet),$$

where, as usual, Q denotes the canonical functor

$$K^-(\text{Ab}) \longrightarrow D^-(\text{Ab}).$$

It follows that for each object $Y^\bullet$ of $D^-(\mathcal{C}_A)$, the functor

$$K^-({}_A\mathcal{C}) \longrightarrow D^-(\text{Ab}), \quad X^\bullet \longmapsto \mathbf{L}\Phi_{X^\bullet}(Y^\bullet)$$

is exact and vanishes on acyclic complexes, hence gives an exact bifunctor

$$D^-({}_A\mathcal{C}) \times D^-(\mathcal{C}_A) \longrightarrow D^-(\text{Ab}),$$

$$(X^\bullet, Y^\bullet) \longmapsto X^\bullet \otimes_A^{\mathbf{L}} Y^\bullet.$$

Passing to homology, one defines the $\text{Tor}_i^A(X^\bullet, Y^\bullet)$ (cf. also [1], 6.3). In the particular case of a commutative ring A , one obtains in fact an exact bifunctor

$$D^-(\mathcal{C}) \times D^-(\mathcal{C}) \longrightarrow D^-(\mathcal{C}),$$

the tensor product in the sense of derived categories ($\mathcal{C} = {}_A\mathcal{C} = \mathcal{C}_A$).

Definition. We shall say that the complex X of $D(\mathcal{C}_A)$ has finite Tor-dimension if it lies in D^- and if there exists $n_o \in \mathbb{Z}^+$ such that

$$\text{Tor}_n^A(X, Y) = 0$$

for every $Y \in \text{ob}({}_A\mathcal{C})$ and every $n > n_o$.

We shall use the notion of Tor-dimension to obtain a characterization of perfect complexes.

Proposition 8.1. Let A be a not necessarily commutative ring and let $L^\bullet$ be a complex of A -modules. The following conditions are equivalent:

- a) $L^\bullet$ is perfect;
- b) $L^\bullet$ is pseudo-coherent, has bounded cohomology, and has finite Tor-dimension.³

For the proof, one uses the following lemma.

Lemma. (Bourbaki, Alg. Comm. I, §2, ex. 15). Every A -module E which is flat and of finite presentation is projective.

Proof of the lemma. Let $u : F \rightarrow F'$ be an epimorphism. We must prove that

$$\text{Hom}_A(E, F) \longrightarrow \text{Hom}_A(E, F')$$

is an epimorphism. It is enough to prove that, for every divisible group G ,

$$\text{Hom}_{\mathbb{Z}}(\text{Hom}_A(E, F'), G) \longrightarrow \text{Hom}_{\mathbb{Z}}(\text{Hom}_A(E, F), G)$$

³cf. SGA 6 I 5.8.1.

is a monomorphism. But the diagram

$$\begin{array}{ccc} \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_A(E, F'), G) & \longrightarrow & \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_A(E, F), G) \\ \sim \downarrow & & \downarrow \sim \\ \mathrm{Hom}_{\mathbb{Z}}(F', G) \otimes_A E & \longrightarrow & \mathrm{Hom}_{\mathbb{Z}}(F, G) \otimes_A E \end{array}$$

is commutative, and the vertical arrows are isomorphisms, by the hypothesis on E .

Proof of Proposition 8.1. The implication a) $\Rightarrow$ b) is immediate. To prove b) $\Rightarrow$ a), let

$$\cdots \longrightarrow P_n \xrightarrow{d_n} \cdots \longrightarrow P_i \longrightarrow 0 \longrightarrow \cdots$$

be a complex isomorphic to $L^\bullet$ in $D(\mathcal{C})$, where the P_j are projective A -modules of finite type. Since $L^\bullet$ has finite Tor-dimension and bounded cohomology, it follows that there exists $n_o \in \mathbb{Z}$ such that:

I) The sequence

$$\cdots \longrightarrow P_{n_o-p} \xrightarrow{d_{n_o-p}} \cdots \longrightarrow P_{n_o} \longrightarrow \mathrm{Ker} d_{n_o+1} \longrightarrow 0$$

is exact.

II) $\mathrm{Ker} d_{n_o+1}$ is flat.

III) The complex

$$0 \longrightarrow \mathrm{Ker} d_{n_o+1} \longrightarrow P_{n_o+1} \longrightarrow \cdots \longrightarrow P_i \longrightarrow 0 \longrightarrow \cdots$$

is isomorphic to $L^\bullet$ in $D(\mathcal{C})$.

But $\mathrm{Ker} d_{n_o+1}$ is of finite presentation, hence projective of finite type by the lemma, which proves 8.1.

If A is a commutative ring, and if $X^\bullet, Y^\bullet$ are two perfect complexes, then their tensor product

$$X^\bullet \overset{\mathbf{L}}{\otimes}_A Y^\bullet$$

in the derived category is again a perfect complex. We obtain a bifunctor

$$D(A)_{\mathrm{parf}} \times D(A)_{\mathrm{parf}} \longrightarrow D(A)_{\mathrm{parf}}$$

which is exact in each variable. Consequently, one defines a product on $K'(A)$, and it is easy to see that $K'(A)$ is thereby endowed with the structure of a commutative ring with unit. In fact, $K'(A)$ can be endowed with the structure of a λ -ring; for details, see the seminar on Riemann-Roch following this one.⁴

The tensor product

$$X^\bullet \overset{\mathbf{L}}{\otimes}_A Y^\bullet$$

of a perfect complex $X^\bullet$ and a bounded pseudo-coherent complex $Y^\bullet$ is a bounded pseudo-coherent complex (A still a commutative ring). Consequently, one can introduce on $K_\bullet(A)$ the structure of a $K'(A)$ -module.

⁴SGA 6 V 2.

9. Functorial properties

a) Let A and B be two not necessarily commutative rings, and let $u : A \rightarrow B$ be a homomorphism. Every left (resp. right) B -module M can be endowed with the structure of a left (resp. right) A -module, denoted $M_{[u]}$. In particular, $B_{[u]}$ is a right A -module.

The functor

$$E \longmapsto B_{[u]} \otimes_A E$$

allows one to define the exact functor

$$D(A)_{\text{parf}} \longrightarrow D(B)_{\text{parf}}$$

and consequently the homomorphism

$$K'(A) \xrightarrow{u^*} K'(B).$$

Thus one obtains a covariant functor from the category of rings (resp. commutative rings) to the category of abelian groups (resp. commutative rings).

b) Suppose now that the left A -module $B_{[u]}$ is pseudo-coherent. It follows that the functor

$$F \longmapsto F_{[u]}$$

transforms a free B -module of finite type into a pseudo-coherent A -module. By using Proposition 5.1, one obtains an exact functor

$$D^b(B)_{\text{coh}} \longrightarrow D^b(A)_{\text{coh}}$$

and consequently a canonical homomorphism

$$K_{\bullet}(B) \xrightarrow{u_*} K_{\bullet}(A).$$

The homomorphisms u_* and u^* satisfy the following identity, the projection formula:

$$u_*(u^*(\alpha) \cdot \beta) = \alpha(u_*(\beta)) \quad \text{for } \alpha \in K'(A), \beta \in K_{\bullet}(B).$$

This easy formula will be proved in a more general form in the exposés of the seminar on Riemann-Roch (SGA 6 IV 2.11.1.2).

10. Relative construction⁵

Let $u : A \rightarrow B$ be a homomorphism of rings, and let

$$D(B) \longrightarrow D(A)$$

be the associated exact functor. In $D(B)$, we can consider the complexes that are pseudo-coherent relative to A , those of finite Tor-dimension relative to A , and also the complexes perfect relative to A . In particular, it is natural to consider the triangulated subcategories

$$D(B)_{A\text{-parf}}, \quad D^b(B)_{A\text{-coh}}$$

⁵cf. SGA 6 III, IV.

of $D(B)$ generated by the complexes that are perfect, respectively pseudo-coherent and bounded, relative to A . These two triangulated categories allow one to define the following groups:

$$K'(B \text{ rel } A) = K(D(B)_{A\text{-parf}}),$$

$$K_{\bullet}(B \text{ rel } A) = K(D^b(B)_{A\text{-coh}}).$$

Examples. Let G be a group and let Λ be a commutative ring. The homomorphism

$$\Lambda \xrightarrow{\varepsilon} \Lambda[G]$$

therefore allows one to define the groups

$$K'(\Lambda[G] \text{ rel } \Lambda), \quad K_{\bullet}(\Lambda[G] \text{ rel } \Lambda),$$

which will be denoted by

$$R'_{\Lambda}(G), \quad R_{\bullet}^{\Lambda}(G).$$

In fact, $R'_{\Lambda}(G)$ is a commutative ring and $R_{\bullet}^{\Lambda}(G)$ is a module over it.

Remarks.

a) If the ring Λ is not necessarily commutative, one may set, by definition,

$$\Lambda[G] = \Lambda \otimes_{\mathbb{Z}} \mathbb{Z}G$$

and consequently obtains a homomorphism

$$\Lambda \xrightarrow{\varepsilon} \Lambda[G].$$

In this way, one may give a meaning to the groups $R'_{\Lambda}(G)$ and $R_{\bullet}^{\Lambda}(G)$ even when Λ is non-commutative.

b) The preceding construction can be generalized to the case of a group prescheme, or of a pro-prescheme in groups, G over $\text{Spec } \Lambda$, by starting from the abelian category of Λ -modules on which G operates; this is no longer made explicit in terms of modules over a ring. One then uses the natural inclusion functor into the category of Λ -modules. If Λ is a field, it can be made explicit in terms of modules over a pro-ring, and in the general case, in terms of comodules, at least if G is affine over Λ ; cf. SGA 3 I 4.7.2.

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⁶Published in SGA 4^{1/2}.